



भारत सरकार

नागर विमानन महानिदेशालय

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Directorate General of Civil Aviation

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Installation and maintenance requirements of Communication Navigation and Surveillance (CNS) equipment

Purpose:

Proper siting, timely calibration, maintenance and environmental control are critical to maintain signal integrity, availability, continuity and reliability of all Communication Navigation and Surveillance (CNS) equipment. The purpose of this Advisory Circular is to provide guidance to Air Navigation Service Provider (ANSP) for effective maintenance and siting conditions of Radio Navigational Aids for safe and efficient functioning of the system.

Introduction:

The siting conditions, timely calibration, maintenance and operation within the service life is vital to ensure the integrity, availability, accuracy, and continuity of CNS equipment/systems.

Proper siting helps to mitigate electromagnetic interference (EMI) from external sources, like nearby transmitters or power lines, which can distort or disrupt the navigation signals. Avoiding such interference is crucial to maintaining operational integrity.

Over time, environmental conditions such as weather, vegetation growth or nearby construction projects may change the suitability of the CNS equipment. Regular calibration and maintenance of CNS system ensures that environmental changes do not degrade the system's effectiveness.

Applicability:

This Advisory Circular is applicable to all civil ANSPs responsible for provisioning CNS Services.

I. Siting Requirements for Radio Navigational Aids

Guidelines are given below for siting requirements of critical and sensitive area of ILS and protected area around DVOR, DME, NDB/Locators. Areas around DGPS platform at stations which are used for flight calibration of navigational aids are also to be protected.

1. ILS (Localizer and Glide Path):

1.1. Definition of ILS Critical and Sensitive areas: -

- a) **Critical Area:** -The ILS critical area is an area of defined dimensions about the localizer and glide path antennas where vehicles, including aircraft, are excluded during all ILS operations. The critical area is protected because the presence of vehicles and/or aircraft inside its boundaries will cause unacceptable disturbance to the ILS signal -in-space;

The critical area is intended to protect the ILS signal in space from moving and stopped aircraft and vehicles. Critical area protection is to be enforced during all ILS operations with protection afforded down to at least the Category I decision height. A critical area disturbance would impact all aircraft using the ILS signal at a given time (entire approach).

- b) **Sensitive Area:** - The ILS sensitive area is an area where the parking and/or movement of vehicles, including aircraft, is controlled to prevent the possibility of unacceptable interference to the ILS signal during ILS operations. The sensitive area is protected against interference caused by large moving objects outside the critical area but still normally within the airfield boundary.

From an operational perspective, the sensitive area would ideally protect aircraft operations at least from the Category I decision height down to the runway, and be activated during low visibility conditions only (e.g. Category II and III). A sensitive area disturbance would normally be of a transient nature, and produce a local disturbance affecting a single aircraft only.

1.2. Protection of Critical and Sensitive Area: -

- a) The critical area should be safeguarded through marked boundaries and suitable signage, limiting access to the area or through procedural means if there is a overlap into operational area.
- b) However, if it is not possible to achieve the ideal situation at a particular installation regarding availability of Critical area, then penetration in critical area should be studied by simulation modelling, mitigated, flight inspected and approved by competent authority of ANSP. Comprehensive report shall be submitted to the O/o DGCA.
- c) The height of grass, shrubs & other vegetation shall be restricted within 150mm in the critical area of ILS.

1.3. Dimensions of ILS Critical and Sensitive area:

1.3.1 ILS Localiser Critical and Sensitive Area dimensions

Localizer critical area and sensitive area dimensions are shown in Figure 1.1

a) LOCALIZER CRITICAL AREA DIMENSIONS:

i. Front of LLZ antenna array

The area of 300 m X 120 m in front of LLZ antenna array is critical area for LLZ.

ii. Back of LLZ antenna Array

The area of [(Localizer antenna array aperture + 10 meters both side) × (10m)] behind the LLZ antenna array is also critical area for LLZ.

b) LOCALIZER SENSITIVE AREA DIMENSIONS:

i. There is no sensitive area for ground vehicle of height ≤ 6 meters for CAT-I/II/III Localizer system.

ii. For other vehicles/aircrafts, LLZ sensitive area are given in table-1 below:

CAT	X _s (Distance from LLZ ant in Approach Direction)	Y _s (Distance from RWY center-line)
I	1300 m	90 m
II/III	Localizer antenna to Runway threshold	180m × K

Table – 1. LLZ SENSITIVE AREA

$$K = \sqrt{\frac{\text{localiser to threshold distance}}{3300 \text{ m}}}$$

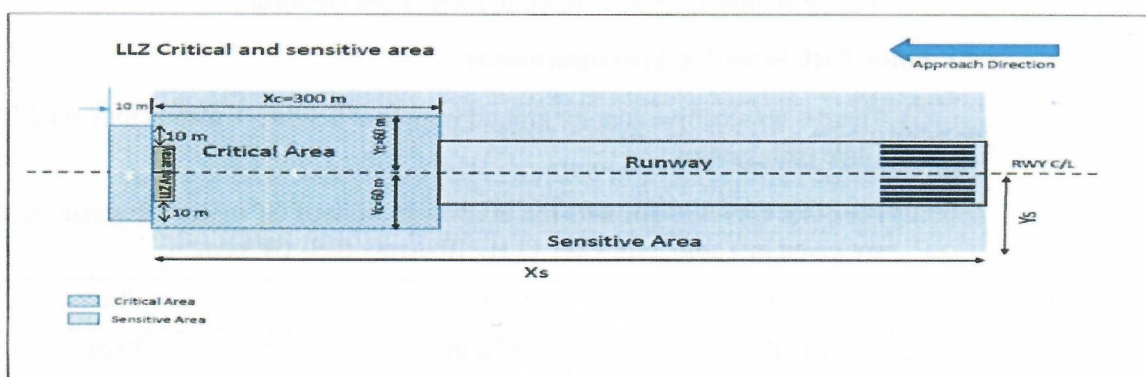


Figure 1.1 . Localizer critical area and sensitive area dimensions

1.3.2 ILS Glide path Critical and Sensitive Area dimensions

Glide path critical area and sensitive area dimensions are shown in Figure 1.2

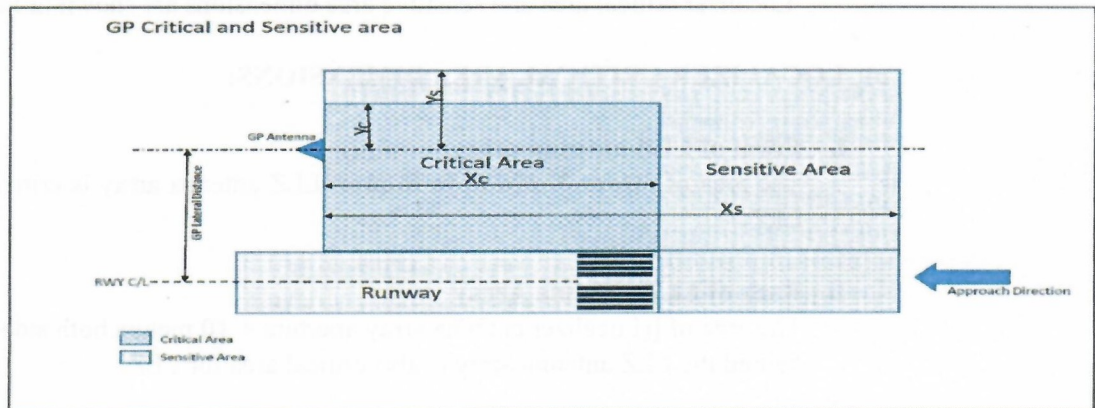


Figure 1.2

a) GLIDE PATH CRITICAL AREA DIMENSIONS:

Critical area dimensions for CAT I / II / III ILS- GP system (M-Array) are given in Table-2 :

Category	Aircraft /vehicles	Ground vehicles heights ≤ 6 m	Ground vehicles / aircraft (on taxiway in front of GP mast) height > 6 m
CAT – I	X_c	299 m or THR distance from GP Ant whichever is higher	610 m
	Y_c	29 m	34 m
CAT – II/III	X_c	299 m or THR distance from GP Ant whichever is higher	672 m
	Y_c	29 m	34 m

Table – 2

X_c : Longitudinal critical distance in meter from GP mast

Y_c : Latitudinal critical distance in meter from GP mast

b) Glide Path Sensitive Area dimensions

- There is no sensitive area for ground vehicle of height ≤ 6 meters for CAT I/II/III Glide path system.
- For other vehicles/ aircrafts (on taxiway in front of GP mast), GP sensitive area are given in Table-3 for CAT I / II / III-Array ILS- GP system:

CAT	X_s	Y_s
I/II/III	672 m	39 m

Table – 3 . GLIDE PATH SENSITIVE AREA

X_s : Longitudinal sensitive distance in meter from GP mast

Y_s : Latitudinal sensitive distance in meter from GP mast

1.4 Technical determination of critical and Sensitive area dimension

If there is any operational limitation for installing LLZ/GP as per the guidelines provided in para 1.3, following guidance should be referred.

In this case a simulation study is mandatory & critical - sensitive area(s) can be redefined for particular airport. However, for all new runways the ILS critical / sensitive areas shall be as per 1.4.1 and 1.4.2.

1.4.1 LOCALIZER CRITICAL & SENSITIVE AREA:

Localizer critical area and sensitive area dimensions are shown in Figure 1.3

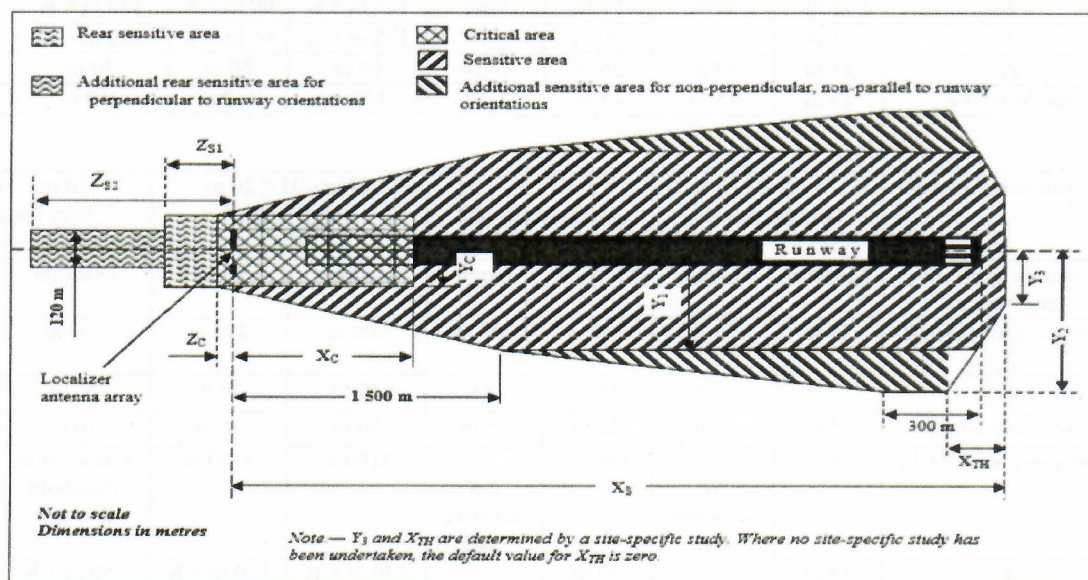


Figure 1.3 Localizer critical and sensitive area dimensions
(Values are as in table below)

Critical & sensitive area dimensions for CAT I / II/ III ILS- LLZ system are given in tables below:

Aircraft/vehicle height	$H \leq 6$ m (see Note 1) Ground vehicle			$6 \text{ m} < H \leq 14$ m Medium aircraft			$14 \text{ m} < H \leq 20$ m Large aircraft		$20 \text{ m} < H \leq 25$ m Very large aircraft	
Antenna aperture (see Note 3)	Small	Medium	Large	Small	Medium	Large	Medium	Large	Medium	Large
Critical area CAT I X_c	180 m	65 m	45 m	360 m	200 m	150 m	500 m	410 m	660 m	580 m
Z_c	10 m	10 m	10 m	35 m	35 m	35 m	50 m	50 m	60 m	60 m
(see Note 10) Y_c	50 m	15 m	20 m	110 m	25 m	25 m	50 m	30 m	55 m	40 m
Sensitive area CAT I X_s	200 m	No sensitive area		500 m	No sensitive area		No sensitive area		1300 m	1100 m
Y_1	40 m			90 m					90 m	50 m
Y_2	40 m			90 m					90 m	50 m
Z_{s1}	15 m			35 m					60 m	60 m
(see Note 7) Z_{s2}	15 m			35 m					60 m	60 m

Aircraft/vehicle height	H ≤ 6 m (see Note 1) Ground vehicle		6 m < H ≤ 14 m Medium aircraft		14 m < H ≤ 20 m Large aircraft		20 m < H ≤ 25 m Very large aircraft	
Antenna aperture (see Note 3)	Medium	Large	Medium	Large	Medium	Large	Medium	Large
Critical area CAT II X_c	75 m	55 m	200 m	200 m	500 m	475 m	750 m	675 m
Z_c	10 m	10 m	35 m	35 m	50 m	50 m	60 m	60 m
(see Note 10) Y_c	15 m	20 m	25 m	25 m	50 m	30 m	70 m	50 m
Sensitive area CAT II X_s	75 m	No sensitive area	500 m	No sensitive area	2100 m	1400 m	Localizer to threshold distance	Localizer to threshold distance
Y_1	15 m		50 m		125 m × K	60 m × K	180 m × K	100 m × K
Y_2	15 m		50 m		125 m × K	60 m × K	180 m × K	125 m × K
Z_{s1}	15 m		35 m		60 m	60 m	70 m	70 m
(see Note 7) Z_{s2}	15 m	15 m	45 m	45 m	160 m	160 m	250 m	250 m

Aircraft/vehicle height	H ≤ 6 m (see Note 1) Ground vehicle		6 m < H ≤ 14 m Medium aircraft		14 m < H ≤ 20 m Large aircraft		20 m < H ≤ 25 m Very large aircraft	
Antenna aperture (see Note 3)	Medium	Large	Medium	Large	Medium	Large	Medium	Large
Critical area CAT III X_c	75 m	55 m	200 m	200 m	500 m	475 m	750 m	675 m
Z_c	10 m	10 m	35 m	35 m	50 m	50 m	60 m	60 m
(see Note 10) Y_c	15 m	20 m	25 m	25 m	50 m	30 m	70 m	50 m
Sensitive area CAT III X_s	100 m	No sensitive area	900 m	No sensitive area	3100 m	3100 m	Localizer to threshold distance	Localizer to threshold distance
Y_1	15 m		50 m		140 m × K	120 m × K	180 m × K	150 m × K
Y_2	15 m		50 m		160 m × K	120 m × K	260 m × K	180 m × K
Z_{s1}	15 m		35 m		60 m	60 m	70 m	70 m
(see Note 7) Z_{s2}	15 m	15 m	45 m	45 m	160 m	160 m	250 m	250 m

X_c and Y_c for an ILS shall depend on type of moving objects (ground vehicles/aircrafts), aperture of antenna (LOC) as per the Note given below.

Notes:

1. For vehicles smaller than 2.5 m in height, $Z_c = 3$ m, assuming a 23-dB front/back ratio for the transmitting antenna for both course and clearance signals.
2. For systems with near-field monitor antennas, vehicles must not enter between the monitor antennas and the transmitting antenna.
3. Small aperture: 11 elements or less. Medium aperture: 12 to 15 elements. Large aperture: 16 elements or more. Simulations have been conducted using a commonly installed 12 element system for the medium and a commonly installed 20 element system for the large aperture cases. It is assumed that Category II/III operations are not conducted on runways equipped with small aperture localizers, and that aircraft as large as a 747 are not operating on such runways.
4. For localizer arrays with very low height, additional critical area will be needed due to the greater attenuation of the direct signal at low vertical angles.
5.
$$K = \sqrt{\frac{\text{localiser to threshold distance}}{3300 \text{ m}}}$$
6. The rear dimensions for sensitive areas may be changed based on specific study results considering fielded antenna pattern characteristics. A directional array with a 23-dB front/back ratio is assumed for course and clearance signals.

7. Single aircraft taxiing or holding parallel to the runway does not generate out-of-tolerance signals.
8. Boundaries for critical areas or rear sensitive areas apply to the entire longitudinal axis (both tail and fuselage) of the interfering aircraft. Boundaries for sensitive areas apply only to the tail of the interfering aircraft.
9. The critical area semi-width, Y_c , should exceed the actual physical dimension of the localizer antenna array by at least 10 m laterally (on both sides) in its portion between the localizer antenna array and the stop end of the runway.

1.4.2 Glide Path (GP) Critical and Sensitive area dimensions: -

Glide path critical area and sensitive area dimensions are shown in Figure 1.4

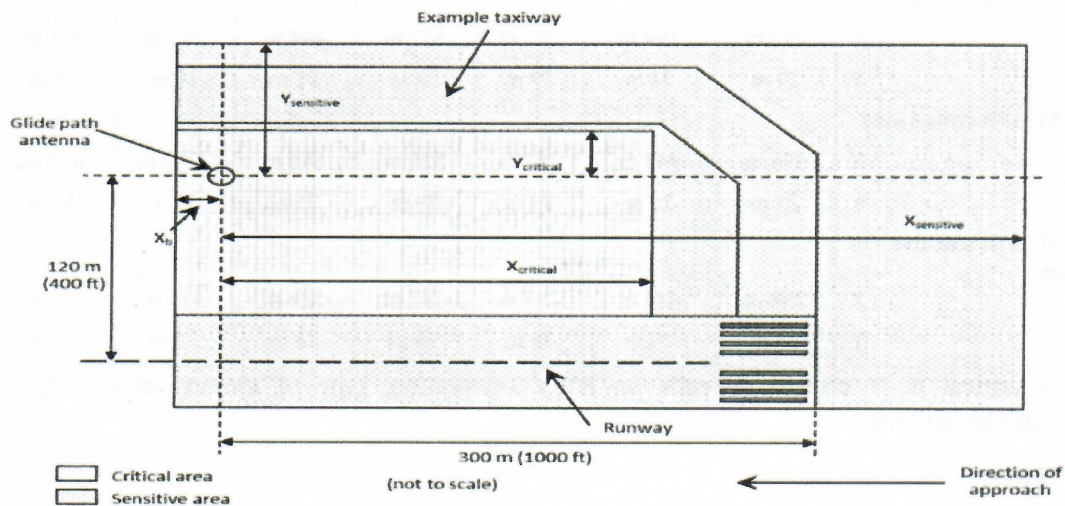


Figure 1.4 Glide path critical and sensitive area dimensions
(Values in associated tables below)

Critical & sensitive area dimensions for CAT I / II / III ILS- GP system for parallel and perpendicular orientations are given in table below:

Aircraft vehicle height	Ground Vehicle		Medium aircraft 6 m < H ≤ 14 m		Large aircraft		Very Large aircraft 20 m < H ≤ 25 m	
Glide path type	M-array	Null-ref	M-array	Null-ref	M-array	Null-ref	M-array	Null-ref
CAT I critical area								
X	299 m	191 m	329 m	829 m	467 m	1117 m	610 m	1360 m
Y	29 m	29 m	20 m	20 m	22 m	22 m	15 m	15 m
CAT I sensitive area								
X	299 m	299 m	270 m	529 m	417 m	717 m	510 m	760 m
Y	29 m	15 m	20 m	20 m	22 m	16 m	15 m	15 m
CAT II/III critical area								
X	299 m	449 m	329 m	829 m	567 m	1267 m	660 m	1410 m
Y	29 m	29 m	20 m	20 m	22 m	22 m	15 m	15 m
CAT II/III sensitive area								
X	299 m	449 m	429 m	629 m	517 m	767 m	560 m	1010 m
Y	29 m	29 m	20 m	20 m	22 m	22 m	15 m	15 m

Critical & sensitive area dimensions for CAT I / II / III ILS- GP system for other orientations are given in following table:

Aircraft vehicle height	Ground Vehicle		Medium aircraft 6 m < H ≤ 14 m		Large aircraft		Very Large aircraft 20 m < H ≤ 25 m	
Glide path type	M-array	Null-ref	M-array	Null-ref	M-array	Null-ref	M-array	Null-ref
CAT I critical area								
X	298 m	191 m	229 m	829 m	444 m	1167 m	591 m	1360 m
Y	24 m	15 m	39 m	39 m	35 m	55 m	34 m	55 m
CAT I sensitive area								
X	298 m	394 m	297 m	537 m	444 m	717 m	541 m	710 m
Y	24 m	24 m	29 m	39 m	25 m	18 m	24 m	24 m
CAT II/III critical area								
X	298 m	443 m	347 m	829 m	544 m	1267 m	672 m	1410 m
Y	24 m	25 m	39 m	39 m	35 m	55 m	34 m	55 m
CAT II/III sensitive area								
X	298 m	445 m	297 m	297 m	528 m	817 m	610 m	1010 m
Y	24 m	24 m	39 m	39 m	25 m	25 m	24 m	24 m

X critical & Y critical depends on RWY orientation, type of moving objects (ground vehicles/aircrafts).

Notes:

1. For $X_b = 50$ m and applies to both critical and sensitive areas for the large and very large aircraft category only. Otherwise, $X_b = 0$ m.
2. The ground vehicle category also applies to small aircraft. Simulations have approximated these aircraft or large ground vehicles using a rectangular box (4 m high × 12 m long × 3 m wide). Depending on local conditions, it may be possible to reduce especially Category I critical area dimensions such that taxiing or driving on the taxiway directly in front of the glide path antenna may be allowed.
3. Separate tables are given for parallel/perpendicular and for other orientations in order to not penalize parallel taxiway operations. To derive worst-case keep-out areas, the largest number among the two tables must be used.
4. Simulations are referenced to the glide path antenna mast using a typical perpendicular distance to the runway centreline of 120 m and a nominal parallel distance from the runway threshold of 300 m. For different antenna-to-runway offsets, the critical and sensitive areas have to be shifted accordingly.
5. The edge of the runway closer to the glide path antenna defines the inner limit of the critical area. The farther edge of the runway defines the inner limit of the sensitive area.
6. Depending on simulation choices (transition point), the critical area may be larger than the sensitive area and impact associated management procedures.
7. Category I critical area is typically equal or larger than the sensitive area, protecting the Category I sensitive area may not be necessary.
8. Boundaries for critical and sensitive areas apply to the entire aircraft (entire fuselage and wings).

2. DVOR: - Protected area & its maintenance: -

- 2.1. DVOR protected area is an area within 300 meters radius from the centre of the antenna of the facility.
- 2.2. Protected area penetration may be approved on a case-by-case basis when necessary to support safe and efficient airfield operations.
- 2.3. The height of grass, shrubs & other vegetation is restricted to be 600 mm in protected area of the DVOR.
- 2.4. Grass cutting equipment or any other vehicle is not to be parked within protected area of the VOR. It can be parked underneath the counterpoise or beyond the protected area of DVOR.

3. NDB/Locators, and DME facilities: -

- 3.1. The height of grass, shrubs & other vegetation is restricted to be 600 mm in protected area of NDB/LOCATOR.
- 3.2. For standalone DME, no structure shall be permitted on land above the level of 3 meters below the antenna base up to a distance of 150 meters from the antenna.
- 3.3. For standalone DME, no steel towers, power lines, metal buildings (located beyond the area of 150M radius) shall protrude in elevation angle of 3 degree measured from the base of Distance Measuring Equipment antenna.

II. Service life requirements

Serviceable life of the CNS equipment/facility should declared by the ANSP using the following factors;

1. Life of Equipment recommended by OEM.
2. Repair support of OEM.
3. Availability of the spares and software in house or marketplace.
4. Performance of facility.
5. Change in technology.

In normal circumstances the CNS facility should not be used beyond the serviceable life. In exceptional cases only use of CNS equipment/facility beyond the expiry of serviceable life may be permitted, on the recommendation of the duly constituted committee of technical experts of that particular equipment/facility. However extension of maximum two years in one step is allowed with an overall capping of five years.

III. Calibration of CNS equipment

Flight calibration schedule of CNS equipment shall be carried out as outlined in following table:

S.No.	Facility	Periodicity
1.	ILS	180 + 30 days
2.	DVOR	720 + 60 days
3.	CVOR	240 + 30 days
4.	Stand Alone DME	720 + 60 days

IV. Maintenance requirements

1. DGPS platform & its maintenance: -

The DGPS platform (used for flight inspections of the navigational facilities of the airport) is to be preserved & maintained properly.

Grass, shrubs and other vegetation near platform should be mown or cut regularly. The height of Grass, shrubs & other vegetation is restricted to be 100 mm around the platform.

2. Signage:

Proper signage shall be provided to delineate the boundaries of critical and sensitive areas of navigational Aids. Sign boards shall be made of non-metallic material and shall be provided as per DGCA CAR Section -4 Aerodrome Design and operations, Series - B Aerodrome Facilities, Part-I Aerodrome design and operations.

3. Water Logging:

There should be no water logging in critical area, sensitive area and around antenna system/equipment shelter of Radio Navigational Aids. Immediate action shall be taken for removal of water, if accumulated due to any reason.

4. Construction of Structures/Obstacles:

Construction of any structure in the vicinity of Navigational Aids are to be controlled as per guidelines issued by Government of India and as amended from time to time. The accountable manager of the airport should take immediate action on unauthorized construction/obstruction which is likely to affect the performance of Radio Navigational Aids .

4.1. No uncovered drainage/water pipe shall be allowed to pass through the critical area of ILS. If drainage is unavoidable in this area, then it should be properly covered & at ground level.

4.2. Electrical power lines above ground level are not permitted to pass through critical area of ILS and protected area in case of DVOR, DME and NDB/Locators.

5. Environmental Control:

The environment in the equipment room shall strictly adhere to the specifications recommended by the Original Equipment Manufacturer (OEM), including Lighting, Temperature, Dust, Seepage & Humidity control and the building/hut should be properly painted.

6. Field Points:

6.1 The Localizer (LLZ) Field Reading Points shall be properly marked and maintained for carrying out measurements/maintenance activities.

6.2 All navigational aid sites, including monitor antennas and field reading points, shall have clear, secure and all-weather access path. These paths must be kept free from vegetation and obstructions to ensure reliable accessibility for maintenance and inspections.

7. Site Protection:

All navigational aid sites shall be protected from unauthorized access or vandalism.

8. Fire Protection:

Fire detection and suppression system shall be installed at navigational sites and the site shall be clear of any fire prone material.

9. Lightning Arrestor:

Lightning arrestor should be installed at navigational sites to protect equipment room and antennas.

10. Earthing System:

The earth resistance of combined earth system shall be less than or equal to 5 ohms. It shall be measured by an calibrated earth testing apparatus for individual earth pits.

11. Obstacle Lights:

Obstacle Lights are required at all the navigational aids sites and antennas as per the guidelines of DGCA Civil Aviation Requirements (CAR) Section 4, Series B, Part I.

12. Media redundancy:

Communication media redundancy should be maintained wherever it is feasible for

a) Remote Control and Monitoring System at the air traffic services unit involved in the maintenance of Navigational aids and

b) Operational status of navigational aids to the unit involved in control of aircraft on the final approach.

13. Power supply:

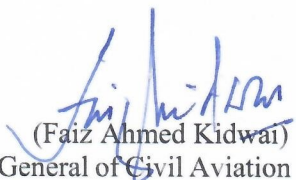
Uninterrupted power supply equipped with surge protection devices shall be provided for navigational aid systems.

V. Change Management:

Major change in siting condition(s) of Navigational Aids shall be submitted to DGCA by ANSP and same will be inspected if required.

VI. Responsibility

The CNS-In-Charge shall be responsible for the upkeep, maintenance and timely calibration of all CNS equipment of the airport.


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